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# Testing by betting when the bets are real: an e-value audit of 84 pre-registered market-efficiency hypotheses

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## Abstract

1 E-values are motivated by a gambler betting against a null hypothesis. We report  
2 an audit in which the gambler is not a metaphor: the null is a price posted by a  
3 bookmaker, the stake is a stake, and the wealth process *is* the e-process. The sub-  
4 ject is a deployed UFC outcome forecaster and a pre-registered search for pockets  
5 where it beats the closing line — 84 hypotheses over 1,787 priced bouts, origi-  
6 nally analysed with paired log-loss and Benjamini–Hochberg. Re-asked as bets,  
7 three things change. (i) The 84 slices are overlapping subsets of one pool, so BH’s  
8 guarantee is not licensed; its one favourable discovery survives neither Benjamini–  
9 Yekutieli nor e-BH. (ii) A segment the fixed- $n$  analysis reported as *failing to repli-*  
10 *cate* ( $p = 0.30$ ) is, on the wealth scale, growing at the same rate in the replication  
11 window and simply 100 bouts short of  $1/\alpha$  — optional continuation converts a  
12 refutation into a schedule. (iii) A post-hoc threshold rule, paid for by a mixture  
13 martingale, survives the search it actually ran ( $e = 31.3$ ) and dies once the direc-  
14 tion it could also have searched is priced in ( $e = 15.6$ ). We report the ladder, not  
15 the winner.

## 16 1 A null hypothesis with money behind it

17 The literature on testing by betting [Shafer, 2021, Grünwald et al., 2024, Ramdas et al., 2023] intro-  
18 duces its central object through a gambler who wagers against a null and whose capital, by construc-  
19 tion, cannot grow in expectation if the null is true. The gambler is usually a device. Here it is the  
20 data-generating situation.

21 We audit VERTEX, a deployed UFC bout-outcome model, against the closing line of a betting ex-  
22 change. The substrate is a walk-forward out-of-fold pool: 4,141 bouts scored between 2016 and  
23 2026 by models refit at 32 quarterly origins, of which 1,787 across 225 events carry an archived  
24 closing price. Two properties of the setting are worth stating because no simulation supplies them.

25 **The null is adversarial and external.**  $H_0$  is not an assumption of convenience — it is a number a  
26 counterparty posted with capital at risk, and a bettor who rejects it collects. Under  $H_0$  the de-vigged  
27 closing probability  $q_t$  is the true conditional probability of the outcome  $y_t \in \{0, 1\}$ .

28 **Predictability is enforced by the calendar.** The model probability  $p_t$  comes from a fit that saw  
29 only bouts strictly before its origin, and  $q_t$  is posted before the fight. Neither has to be argued into  
30 being predictable.

31 The prior analysis of this pool [Prigodskii, 2026] declared 84 segment hypotheses — style matchups,  
32 durability, market microstructure, form, pedigree, division context — in a registry, before scoring  
33 any of them, each with a stated mechanism for why a good market would be wrong in that exact  
34 place. It then measured each with a composition-adjusted paired log-loss contrast, cluster-robust by

event, and charged multiplicity with BH. It also reported its own power: against a 3 percentage-point edge the paired test has 10–15% power, *falling to 0–1% once BH multiplicity is charged*, and would need  $\sim 89,900$  bouts. There are 1,787. That report is the reason for this one: the instrument was known to be inadequate before its answers were read.

## 2 The wealth process

For a bout with de-vigged closing probability  $q$  and outcome  $y$ , and a predictable stake  $\lambda$ ,

$$M = 1 + \lambda(y - q), \quad \lambda \in \left(-\frac{1}{1-q}, \frac{1}{q}\right), \quad \mathbb{E}_{H_0}[M] = 1. \quad (1)$$

The product over a segment’s bouts *in calendar order* is a non-negative test martingale; its terminal value is an e-value and, by Ville’s inequality, its running maximum is anytime-valid. The growth-rate optimal stake under the alternative “the model’s  $p$  is the truth” is available in closed form,

$$\lambda^* = \frac{p - q}{q(1 - q)}, \quad (2)$$

the model’s edge deflated by the market’s own variance. We report fractional Kelly at  $\lambda = \frac{1}{2}\lambda^*$ , and, because that fraction is a free choice, also a predictable plug-in that learns the multiplier from past bouts only (§7).

**The margin is a supermartingale, not an excuse.** Settling the same stakes at the price a better can actually get, rather than at the fair  $1/q$ , gives  $\mathbb{E}_{H_0}[M] = 1 - f(1 - qd) \leq 1$ , since  $qd < 1$  whenever the book charges an overround. The deployable process is therefore a valid e-value that is uniformly conservative, short of the statistical one by exactly what the margin costs. On this pool the mean overround is 1.0482 and the shortfall is 0.00525 nats per bout: a measurement of how much statistical evidence the bookmaker’s cut consumes.

**Validation before use.** Over 40,000 replications under  $H_0$ ,  $\mathbb{E}[e] = 1.000 \pm 0.003$ ;  $\Pr(\sup_t e_t \geq 1/\alpha)$  is 0.021, 0.058 and 0.142 at  $\alpha = 0.05, 0.10, 0.20$ ; e-BH applied to 84 pure nulls rejects nothing; the confidence sequence of §6 covers the truth at *every*  $t$  in 98% of runs.

**Where the model actually stands.** Betting the model against the book over the whole priced pool gives  $e = 3.6 \times 10^{-4}$  at fair odds and  $3.1 \times 10^{-8}$  at real ones. This is not a paper about beating a market. It is a paper about what can be honestly claimed inside a search that mostly failed.

## 3 The multiplicity charge BH was not licensed to make

The 84 slices are overlapping subsets of one pool of 1,787 bouts, scored by one model against one book. A single upset moves dozens of them together, with a dependence structure nobody has characterised. It is also not the benign kind: across the 3,486 segment pairs, membership correlations run from  $-0.49$  to  $+0.84$  with half of them negative, and 83 pairs share no bouts at all. This is not the positively-dependent family for which BH’s guarantee is routine, and PRDS would have to be argued rather than assumed. BH’s FDR guarantee requires independence or PRDS [Benjamini and Yekutieli, 2001]; neither is established here. The classical repair is Benjamini–Yekutieli, which charges  $\sum_{i \leq 84} 1/i = 5.01$ . e-BH [Wang and Ramdas, 2022] is valid under *arbitrary* dependence and charges nothing.

On the discovery window the lab searched (1,191 bouts, the same 84 hypotheses):

| procedure                             | guarantee holds? | rejects |                         |
|---------------------------------------|------------------|---------|-------------------------|
| BH, paired statistic                  | not established  | 4       | 1 favourable, 3 against |
| Benjamini–Yekutieli ( $\times 5.01$ ) | yes              | 0       |                         |
| e-BH, wealth statistic                | yes              | 0       | cutoff $e \geq 1680$    |

The lab’s single favourable discovery — a fighter carrying a win streak of four or more, BH  $q = 0.016$  — does not survive either procedure whose guarantee actually holds. The two valid procedures agree, which is the substantive point; they disagree with BH, which is the correction.

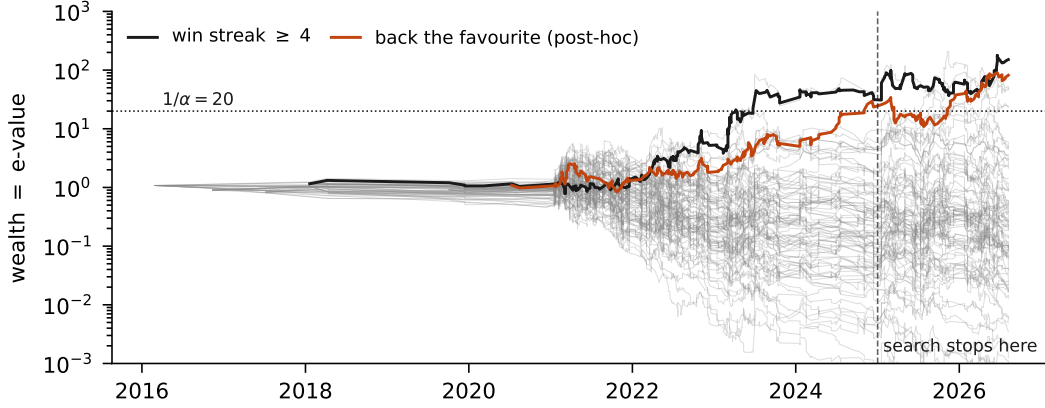


Figure 1: Wealth as a function of calendar time for every pre-registered segment (grey) and the two the analysis singled out (coloured). The dashed line is where the search stopped and the replication window began. A fixed- $n$  analysis must start a new test to the right of it; a test martingale does not.

It would be wrong to read this as e-values buying less. The same segment's e-value on the full pool is 150.9, and for a *single* pre-specified hypothesis  $e \geq 20$  is decisive evidence. What is expensive is not the machinery but the charge for having asked 84 questions of a pool this size — the same lesson the original analysis reached, now in a currency where the arithmetic is visible rather than delegated to a  $q$ -value.

#### 4 “Failed to replicate” and “not yet” are different findings

The original analysis froze its discovery window, selected the streak segment, and ran a fresh fixed- $n$  test on 2025–26. That test returned  $p = 0.30$  and was reported, correctly under its own logic, as a failure to replicate. It is the only move a fixed- $n$  design has: the second window buys a second test and nothing carries over.

On the wealth scale the same 110 bouts read differently. The segment's log-growth in the replication window is  $+0.0143$  nats per bout — statistically indistinguishable from, and in fact slightly above, its discovery-window rate. The e-value accumulated there is 4.81: real evidence, short of  $1/\alpha = 20$ . At that growth rate, crossing needs 210 bouts; 110 have arrived. The finding is not refuted. It is 100 bouts — roughly eighteen months of cards — from a decision, and an e-process lets one simply keep multiplying rather than choose a stopping time in advance (Figure 1). We report the pre-split wealth ( $e = 31.4$ ) but do not claim it: that is the window in which the segment was chosen.

The effect is stable across the five walk-forward seeds the pool carries (e-values 56.2 to 150.9, all above  $1/\alpha$ ), which the  $q$ -value machinery had no natural place to put.

#### 5 Pricing the freedom a post-hoc rule used

The largest effect in the original analysis is not a segment but a direction: where the model gives the book's favourite *more* than the book does, flat-staking that favourite returned  $+11.5\%$  over 281 bets across 163 events. The analysis declared this post-hoc and said so — the cut at lean  $\geq 0.05$  was chosen after seeing that the sweep was positive from 0.02 to 0.22 and peaked at 0.13. A fixed- $n$  framework has no principled penalty for that.

A mixture martingale has one, and it is a price fixed in advance rather than argued after: each threshold's wealth process is a martingale starting at 1, so any convex combination is too, and the search is paid for by the prior mass not placed on the winner. The maximum over thresholds is 118.99 and is not a valid e-value. The uniform mixture over the 21 cuts is  $e = 31.3$ , which clears  $1/\alpha$ . Widening the family to the 21 cuts the search could also have taken in the opposite direction gives  $e = 15.6$ , which does not. Adding the other disagreement statistic available on the same frame gives 10.4.

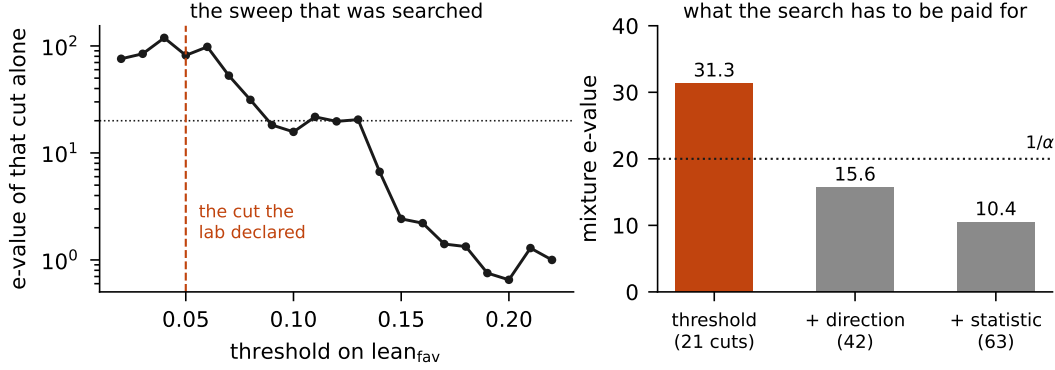


Figure 2: Left: the e-value of each threshold on the model’s lean toward the book’s favourite. The declared cut at 0.05 was chosen after this shape was visible. Right: a mixture martingale over progressively more of the freedom the search actually had. The finding survives paying for the threshold and does not survive paying for the direction.

106 We regard the ladder, not any rung, as the result. It converts “this was post-hoc” from a caveat into a  
 107 quantity: this finding can absorb the threshold search and cannot absorb the threshold search plus the  
 108 choice of direction. We are not aware of a fixed- $n$  analysis that can state the boundary that precisely.

## 109 6 What the sequential instrument needs

110 Replacing the minimum detectable effect with a growth rate makes project planning a calculation  
 111 rather than a verdict. Against a true 3 pp edge simulated on this pool’s own price distribution, the  
 112 half-Kelly bet grows at 0.00166 nats per bout and reaches  $e = 1/\alpha$  in  $\sim 1,800$  bouts; the paired  
 113 log-loss test needs  $\sim 5,346$  at 80% power. At 5 pp the figures are 648 and 1,902. The sequential  
 114 test is roughly  $3\times$  cheaper in fights, not because it is more powerful per observation, but because a  
 115 paired contrast on a slice discards most of what each bout carries.

116 Anytime-validity is not free, and the honest comparison says so. For the directional rule, the cluster  
 117 bootstrap gives a 95% interval on ROI of  $[+2.2\%, +20.4\%]$  — valid at the one  $n$  at which the  
 118 analysis stopped, which it chose after watching the number. A hedged betting confidence sequence,  
 119 valid at every  $n$  simultaneously, gives  $[-5.3\%, +22.9\%]$  and does not exclude zero.

## 120 7 Limitations

121 The stake is a free choice. Fractional Kelly at 0.25/0.5/1.0 gives  $e = 26.0/150.9/115.3$  on the  
 122 streak segment, and a predictable plug-in that learns the multiplier from past bouts only — so not a  
 123 choice at all — settles at  $c = 0.70$  and  $e = 63.1$ . Every variant clears  $1/\alpha$ ; the conclusion is not the  
 124 fraction.

125 Boosting e-BH [Wang and Ramdas, 2022] would recover power, but it requires the null distribution  
 126 of each e-value, which a composite martingale null does not supply. Pre-specified weights across  
 127 the nine declared families are a better route and were not pre-specified.

128 Three contaminations are inherited from the underlying pool and were sized there: the served model  
 129 carries an age correction whose shipping decision used data inside the replication window; the  
 130 model’s hyperparameters were tuned on a window the pool later scores (122 of 1,191 discovery  
 131 bouts); and one derived rating is not fully point-in-time, contaminating 21.9% of pool bouts, worth  
 132 0.0011 nats. Each is small and each moves the reported numbers in the unfavourable direction if  
 133 removed.

134 Finally, this is one sport, one book, and one model. The construction generalises to any forecaster  
 135 audited against a market price; the numbers do not.

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